



Youssef FELLAH

Software Engineering Student & Systems Architecture

Seeking Final-Year Internship (PFE) - Formal Design, Distributed Systems & Cloud

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PROFESSIONAL PROFILE

Software Engineering student at Université Mundiapolis, specializing in the formal analysis of business requirements, architectural modeling (UML, BPMN, Clean Architecture), and distributed systems design. Complementary experience in AI integration (RAG/LLM), ERP, and blockchain/Web3 development. Rigorous and focused on durable architecture, I am seeking a final-year internship (PFE) to lead the study, modeling, and sizing of modular, secure, and highly scalable software solutions.

PROFESSIONAL EXPERIENCE

BROKER IMMOBILIER

Casablanca, Morocco

Software Architect & Systems Engineer — SaaS & AI Infrastructure (Work-Study)

Feb. - May 2026

- Modeled and architected the foundation of a multi-tenant SaaS CRM for the real estate sector following Clean Architecture principles (Spring Boot, Angular), ensuring strict isolation of the business domain.
- Designed and deployed a RAG (Retrieval-Augmented Generation) architecture and LLM integrations for semantic search and automated summarization of real estate documents.
- Secured application flows by implementing granular role-based access control (RBAC) and a stateless JWT authentication mechanism.

HTBS AFRICA

Casablanca, Morocco

Information Systems & ERP Engineering Intern — Finance

Aug. - Oct. 2025

- Analyzed and translated Moroccan tax and treasury regulations into formal technical specifications for adapting the Dolibarr ERP.
- Designed and integrated 4 custom extension modules, automating critical business processes without altering the application core.
- Optimized the relational schema and refactored SQL queries, cutting accounting and tax processing execution time by 40%.

FST ERRACHIDIA — Final-Year Project (Bachelor's PFE)

Blockchain Developer — Web3

Apr. - Jul. 2025

- Designed and deployed a decentralized e-learning platform on Ethereum, combining on-chain smart contracts with IPFS-distributed content storage.
- Authored Solidity smart contracts for verifiable certification and built a React frontend with Web3 wallet interactions.

MAJOR ENGINEERING PROJECTS

Business Modeling & ERP Architecture (Odoo) — BPMN, Python

- Formalized logistics flows for perishable goods using the BPMN standard to guarantee traceability and alignment with operational constraints; extended the framework via the Open-Closed principle (GoF).

Distributed Architecture: Core Banking — Java, Spring Boot, PostgreSQL

- Designed a decoupled architecture rigorously isolating the central banking transaction engine from external interaction layers, with a secure integration bus under RBAC authorization.

devctl — Platform Engineering & Local Orchestration — Python, Docker, Bash

- Specified and automated a unified CLI for orchestrating Spring Boot, Angular, and Vue.js environments through a single command surface, eliminating cross-environment inconsistencies.

Decentralized Edu — Solidity, React.js, Web3.js

- Blockchain-native e-learning platform using smart contracts for tamper-proof certification issuance and IPFS for resilient content storage.

Infrastructure Supervision & Firewall Config — AWS, Zabbix, OPNsense

- Cloud monitoring with Zabbix on AWS (Linux/Windows); OPNsense firewall deployment with VLAN segmentation and enterprise-grade network hardening.

Software Quality & Engineering Governance — Git, CI/CD, Documentation

- Defined the branching strategy and continuous validation policies; authored architecture guidelines and provided technical mentorship to development teams.

TECHNICAL ARSENAL

Modeling & Architecture	UML, BPMN, Clean Architecture, Design Patterns (GoF), DDD, REST & RPC APIs
Languages & Frameworks	Java (Spring Boot, Security), Python, PHP, Node.js, TypeScript, Solidity
Frontend	Angular, React.js, Flutter, JavaScript, HTML5/SCSS, Tailwind, Web3.js
Blockchain & Web3	Decentralized protocols, smart contracts, Solidity, IPFS, Ethereum
Cloud & Infra.	Huawei Cloud, AWS (EC2, S3, RDS), Docker, CI/CD, Linux, Windows Server, OpenStack, Zabbix, Shell
Data & AI	RAG Pipelines, LLM APIs, Vector DBs, PostgreSQL, MySQL, MongoDB, Hibernate/JPA, SQL Optimization, Power BI, Tableau
Security & Networks	JWT Authentication, RBAC Modeling, SOLID Principles, OPNsense, Firewall, IoT, Endpoint Security

CERTIFICATIONS & EDUCATION

- Huawei: HCIA-Cloud Computing V5.5
- AWS Academy: Cloud Foundations
- IBM · Coursera: Introduction to Software Engineering
- GitHub: Foundations (Parts 1 & 2) and Admin & Product Features
- Cisco: Data Analytics Essentials (2026)
- Packt · Coursera: Advanced Java APIs & Functional Programming

Engineering Degree in Software Engineering
Université Mundiapolis, Casablanca · 2025 - Present

Bachelor's Degree (LST) in Software Engineering
FST Errachidia, Morocco · 2024 - 2025

Languages
French & Arabic (native) · English (Technical / C1)